

# Subspace Guessing and Rank-Metric Solvers with Hints

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## Abstract

We show how to improve rank-metric solvers when certain side information (hints) about the secret is available. Concretely, we adapt the kernel search algorithm for MinRank and the GRS algorithm for the Rank Syndrome Decoding problem when some entries in the rank decomposition of the error matrix are known. This setting is motivated by side-channel leakage and cryptographic applications: Mirath and RYDE, two signature candidates in the NIST post-quantum competition, rely on these problems and employ secret keys in this decomposed form. As a main technical ingredient, we give an optimal procedure for guessing a subspace containing the row space of a systematic matrix given only partial knowledge of its entries. Further, we describe a profiling side-channel attack on the reference implementation of Mirath to demonstrate the plausibility of obtaining such hints.

**Keywords:** rank-metric, coding theory, cryptography, side-channel.

## 1 Introduction

The MinRank and the Rank Syndrome Decoding (RSD) problems form the basis of the security of many recently proposed cryptographic schemes and, in particular, of Mirath [AAB<sup>+</sup>25] and RYDE [ABB<sup>+</sup>25] which are two digital signature candidates in the current NIST post-quantum standardization process [ABC<sup>+</sup>24]. Understanding the complexity of solving these problems is integral to determining the security level of these schemes. More generally, there is significant interest in understanding how this complexity changes when partial information (*hints*) about the secret becomes available, for instance through side-channel leakage.

These leakage settings have been extensively studied in cryptography, both for classical schemes (e.g., RSA and Diffie–Hellman), see [MH24], and for post-quantum schemes, see, e.g., [EMVW22]. These works typically focus on bit-level hints (also called *partial key exposure* attacks), i.e., knowledge of some random bits of the secret. Other forms of hints that are not purely bit-level have also been considered, for example by Dachman-Soled et al. [DSDGR20] for lattice problems and by Horlemann et al. [HPR<sup>+</sup>22] for the Syndrome Decoding problem in the Hamming metric. As an example, the latter considers hints such as knowing the  $\mathbb{F}_q$ -entries of the error vector or weights of its subvectors.

In both MinRank and RSD, the secret can be represented as an  $m \times n$  matrix  $\mathbf{E}$  over  $\mathbb{F}_q$  of small rank  $r$ . In cryptographic implementations,  $\mathbf{E}$  is often represented via a rank decomposition with a systematic part which is more efficient in terms of key size, for example:

$$\mathbf{E} = \mathbf{S}(\mathbf{I}_r \mid \mathbf{C}') \text{ where } \mathbf{S} \in \mathbb{F}_q^{m \times r}, \mathbf{C}' \in \mathbb{F}_q^{r \times (n-r)}.$$

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Table 1: Rank-Metric Hints Considered in the Literature

| Work <sup>a</sup> | Problem | Hint <sup>b</sup>                                                                                                                                                                                                                                  | Method              | Motivation                                                                                                                                                                            |
|-------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [DEGS25]          | RSD     | Bits of $\mathbf{e} \in \mathbb{F}_{2^\nu}^n$                                                                                                                                                                                                      | Adapt GRS           | Older RYDE versions used $\mathbf{e}$ as secret key                                                                                                                                   |
| [DEGS25]          | RSD     | Bits of the coefficients of the $q$ -polynomial that annihilates $\mathbf{e} \in \mathbb{F}_{2^\nu}^n$                                                                                                                                             | Reduce to MinRank   | Older RYDE versions used this $q$ -polynomial as the signing witness                                                                                                                  |
| [DEGS25]          | MinRank | Bits of $\mathbf{x} \in \mathbb{F}_2^n$                                                                                                                                                                                                            | Reduce to MinRank   | MIRA and MiRiTH (older NIST candidates) used $\mathbf{x}$ as secret key                                                                                                               |
| [DEGS25]          | MinRank | Bits of the coefficients of the $q$ -polynomial that annihilates $\mathbf{E} \in \mathbb{F}_2^{m \times n}$ when $\mathbf{E}$ is represented as an element of $\mathbb{F}_{2^\nu}^n$                                                               | Reduce to MinRank   | MIRA used this $q$ -polynomial as the signing witness                                                                                                                                 |
| <b>This work</b>  | MinRank | $\mathbb{F}_q$ -entries or bits of $\mathbf{S}$ and $\mathbf{C}'$ in $\mathbf{E} = \mathbf{S}(\mathbf{I}_r \mid \mathbf{C}') \in \mathbb{F}_q^{m \times n}$                                                                                        | Adapt kernel search | Mirath uses $(\mathbf{S}, \mathbf{C}')$ as secret key                                                                                                                                 |
| <b>This work</b>  | RSD     | $\mathbb{F}_q$ -entries or bits of $\mathbf{S}$ and $\mathbf{C}'$ in $\mathbf{E} = \mathbf{S}(\mathbf{I}_r \mid \mathbf{C}') \in \mathbb{F}_q^{m \times n}$ where $\mathbf{E}$ is the matrix representation of $\mathbf{e} \in \mathbb{F}_{q^m}^n$ | Adapt GRS           | RYDE uses $(\mathbf{s}, \mathbf{C}')$ as secret key where <sup>c</sup> $\mathbf{s} \in \mathbb{F}_{q^m}^r$ is the vector representation of $\mathbf{S} \in \mathbb{F}_q^{m \times r}$ |

<sup>a</sup>The work [DEGS25] also describe the *error* model, where some hints may be corrupted, unlike the *erasure* model, where hints are always correct (as in our work). Their proposed algorithm is to guess which hints are correct and then run the erasure-model solver, thus effectively reducing the problem to the erasure setting.

<sup>b</sup>See Sections 3 and 4 for definitions of  $\mathbf{e}$ ,  $\mathbf{x}$ , etc.

<sup>c</sup>In addition to choosing  $\mathbf{C}$  systematic, RYDE also specializes the first entry of  $\mathbf{s}$  to 1.

In this work, the hints consist of partial knowledge of the  $\mathbb{F}_q$ -entries of  $\mathbf{S}$  and  $\mathbf{C}'$ . We also consider the *bit-level* variants, i.e., knowledge of some bits of some  $\mathbb{F}_q$ -entries (when  $q = 2^\nu$ ). These choices are motivated by the fact that the secret key in both Mirath and RYDE is of the form  $(\mathbf{S}, \mathbf{C}') \in \mathbb{F}_q^{m \times r} \times \mathbb{F}_q^{r \times (n-r)}$ , making it plausible for a side-channel attack to reveal such information.

Concretely, we adapt the kernel search [GC00] solver for MinRank and the GRS solver [GRS16] for the RSD problem under the hints model mentioned above. These two combinatorial algorithms form one of the two main classes of rank-metric solvers for cryptographically relevant parameters (the other being the algebraic solvers of [BBC<sup>+</sup>20, BBB<sup>+</sup>23]).

The only prior work to our knowledge that considers rank-metric hints is the recent work of D’Alconzo et al. [DEGS25]. All the hints analyzed by them are bit-level. We list them alongside our hints in Table 1 for easier comparison.

## A Contribution

### 1) Subspace Guessing with Hints (Section 2)

As the main building block (Theorem 1), we solve the following problem, which may be of independent interest: Given a random systematic matrix  $\mathbf{C} := (\mathbf{I}_r \mid \mathbf{C}') \in \mathbb{F}_q^{r \times n}$  and  $\ell > 0$  with some entries of  $\mathbf{C}'$  unknown, find a subspace of dimension  $r + \ell$  that contains the row space of  $\mathbf{C}$  with highest probability. For

example,  $\mathbf{C}$  might be

$$\left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & * & 2 & * & * \\ 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 1 & * & 1 & * & * & * \end{array} \right] \in \mathbb{F}_3^{3 \times 8},$$

where the  $*$  represent unknown entries.

## 2) Kernel Search and GRS with Hints (Sections 3 and 4)

The MinRank problem asks to find a matrix  $\mathbf{E}$  of small rank  $r$  in a space spanned by  $K$  matrices in  $\mathbb{F}_q^{m \times n}$ . Then the complexities of our modified kernel search algorithm with hints and (for comparison) the original kernel search are

$$\tilde{K}^\omega q^{r[\tilde{K}/m] - h_c + t_c} \quad \text{and} \quad K^\omega q^{rK/m}$$

respectively. Here  $h_s$  denotes the number of known entries in  $\mathbf{S}$ ,  $\tilde{K} := \max\{0, K - h_s\}$ ,  $h_c$  is the number of known entries in  $\mathbf{C}'$  and  $t_c$  is the number of known entries within any  $\ell := n - r - \lceil \tilde{K}/m \rceil$  columns of  $\mathbf{C}'$  with the fewest hints. Section 3 describes the complexity for bit-level hints.

The RSD problem asks to find an error  $\mathbf{e} \in \mathbb{F}_{q^m}^n$  of small rank  $r$  over  $\mathbb{F}_q$  that satisfies a certain parity-check equation. This error can also be represented as a matrix  $\mathbf{E} \in \mathbb{F}_q^{m \times n}$  of rank  $r$ . The complexities of our modified GRS algorithm (for the “row space” version) and (for comparison) the original approach are

$$(n - k)^\omega m^\omega \max\{q^{r(k - \lfloor h_s/m \rfloor) - h_c + t_c}, 1\}$$

and  $(n - k)^\omega m^\omega q^{rk}$

respectively, where  $h_s$ ,  $h_c$  and  $t_c$  have similar meanings as in the previous part but with  $\ell := n - k - r + \lfloor h_s/m \rfloor$ . See Section 4 for the complexities of the “column-space” version and bit-level hints.

**Limitations** Both kernel search and GRS can guess on either  $\mathbf{C}$  or  $\mathbf{S}$ . When  $m = n$  this choice does not matter, but for other parameter sets it can affect the complexity. However, our method of incorporating hints only allows using the matrix that is in systematic form which may not be optimal depending on  $m$  and  $n$ . (Although we note that cryptographic schemes typically use  $m = n$ .) Moreover, we adapt only the original GRS algorithm [GRS16], and not the improvement in [AGHT06], which performs better for certain parameter regimes. For details, see Section 6.

**Concrete Cost Examples** Table 2 shows the complexity in bits of the modified kernel search and GRS (row-space) with hints algorithm respectively, where the guessing is done on the right (i.e., we guess the row space of  $\mathbf{C}$ ). Results are shown for the **Mirath-1a** parameter set ( $q = 16, m = 16, n = 16, K = 143, r = 4$ ) and the **RYDE-1** parameter set ( $q = 2, m = 53, n = 53, k = 45, r = 4$ ), both targeting 143-bit classical security, corresponding to NIST Category I (AES-128). We assume each entry of  $\mathbf{C}'$  (resp.  $\mathbf{S}$ ) is revealed with probability  $h_{c,\text{prob}}$  (resp.  $h_{s,\text{prob}}$ ). The tables show how the complexity decreases as  $h_{c,\text{prob}}$  increases from 0.0 to 1.0 while  $h_{s,\text{prob}} = 0$  and vice versa. Note the  $h_{c,\text{prob}} = 0$  and  $h_{s,\text{prob}} = 0$  case correspond to the complexity of the original algorithms without hints. We set  $\omega = 2.81$ , which is what Mirath assumes. The costs were computed as  $\log_2(P \log_2(q)/S)$  where  $P$  is the average polynomial cost in  $\mathbb{F}_q$ -operations and  $S$  the average success probability, with the average taken over  $10^6$  samples over the randomness of  $h_s$ ,  $h_c$  and  $t_c$ .

## 3) Side-Channel Implementation (Section 5)

To show that this hints setting is plausible, we performed a profiling attack on Mirath [AAB<sup>+</sup>25] that recovers the entries of  $\mathbf{C}$  with high probability. The attack only targets the reference implementation (no optimized versions were available) and standard countermeasures such as masking would prevent it. Still, together with our theoretical results, this highlights the need for side-channel countermeasures in rank-metric schemes.

Table 2: Concrete Complexity (in bits) of Kernel Search and GRS with Hints

| Scheme+Solver <sup>a</sup> | Hint Prob. <sup>b</sup> | 0.0 | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 | 1.0 <sup>c</sup> |
|----------------------------|-------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------------------|
| Mirath-1a+KS-H             | $h_{c,\text{prob}}$     | 166 | 104 | 72  | 46  | 32  | 26  | 23  | 22  | 22  | 22  | 22               |
| Mirath-1a+KS-H             | $h_{s,\text{prob}}$     | 166 | 159 | 152 | 143 | 137 | 130 | 123 | 118 | 116 | 107 | 100              |
| RYDE-1+GRS-H               | $h_{c,\text{prob}}$     | 205 | 176 | 150 | 124 | 104 | 84  | 66  | 49  | 33  | 25  | 25               |
| RYDE-1+GRS-H               | $h_{s,\text{prob}}$     | 205 | 205 | 204 | 201 | 200 | 197 | 197 | 196 | 193 | 193 | 189              |

<sup>a</sup>The Mirath-1a and RYDE-1 parameter sets both target 143-bit classical security. KS-H and GRS-H denote the modified kernel search and GRS (row-space version) with hints solver respectively. Note: RYDE specializes the first entry of  $\mathbf{s}$  to 1, but we do not exploit this to compute the costs here. To do so, one would increase  $h_s$  by  $m$ , since this reveals the first column of  $\mathbf{S} \in \mathbb{F}_q^{m \times r}$ .

<sup>b</sup>When only  $h_{c,\text{prob}}$  or  $h_{s,\text{prob}}$  appears, the other probability parameter is implicitly 0.

<sup>c</sup>When  $h_{c,\text{prob}} = 1.0$  the complexity is only polynomial (this already occurs at 0.6 for Mirath). When  $h_{s,\text{prob}} = 1.0$  the exponential term still remains since we are guessing the row space of  $\mathbf{C}$  which is unknown. Of course, in the latter, one can recover the secret efficiently since  $\mathbf{S}$  is entirely known.

## B Organization

Section 2 shows (Theorem 1)) how to solve the “subspace guessing with hints” problem. Section 3 defines the MinRank problem and then, using this result, shows how to incorporate hints into the kernel search solver. Section 4 defines the RSD problem and then does the same for the GRS solver. Section 5 describes our side-channel profiling attack. Section 6 concludes with future work.

## C Notation

We assume row vectors. Let  $\mathbf{A} \in \mathbb{F}_q^{m \times n}$  be an  $m \times n$  matrix over the finite field  $\mathbb{F}_q$ . Then  $A_{i,j} \in \mathbb{F}_q$  is the entry in row  $i$  and column  $j$ . Denote by  $[a : b]$  the set of integers from  $a$  to  $b$ , inclusive, and let  $[b]$  be shorthand for  $[1 : b]$ . If  $\mathcal{I} \subseteq [m]$  and  $\mathcal{J} \subseteq [n]$ , then  $\mathbf{A}_{\mathcal{I},\mathcal{J}} \in \mathbb{F}_q^{|\mathcal{I}| \times |\mathcal{J}|}$  is the submatrix of  $\mathbf{A}$  with rows and columns indexed by  $\mathcal{I}$  and  $\mathcal{J}$  respectively. The notation  $\mathbf{A}_{*,\mathcal{J}} \in \mathbb{F}_q^{m \times |\mathcal{J}|}$  denotes the submatrix consisting of columns indexed by  $\mathcal{J}$  and all rows, and analogously for  $\mathbf{A}_{\mathcal{I},*}$ . We sometimes write  $\mathbf{I}_m$  or  $\mathbf{0}_{m \times n}$  to make the size of the identity or zero matrix clear. The matrix multiplication exponent is  $\omega$ .

## 2 The Subspace Guessing with Hints Problem

**Proposition 1** (Optimal Subspace). *Let  $\mathbf{C} := (\mathbf{I}_r \mid \mathbf{C}') \in \mathbb{F}_q^{r \times n}$  be a matrix in systematic form, where only some entries of  $\mathbf{C}' \in \mathbb{F}_q^{r \times (n-r)}$  are known, and let  $\ell > 0$ . Let  $\mathcal{C}$  be the set of all possible matrices  $(\mathbf{I}_r \mid \mathbf{C}'') \in \mathbb{F}_q^{r \times n}$  such that  $\mathbf{C}'' \in \mathbb{F}_q^{r \times (n-r)}$  agrees with  $\mathbf{C}'$  at the known entries. Then there exists an  $(r + \ell)$ -dimensional subspace  $\mathcal{R} \subseteq \mathbb{F}_q^n$  such that*

$$|\{ (\mathbf{I}_r \mid \mathbf{C}'') \in \mathcal{C} : \text{rowspan}(\mathbf{I}_r \mid \mathbf{C}'') \subseteq \mathcal{R} \}| = q^{r\ell - t_c}$$

and for no other  $(r + \ell)$ -dimensional  $\mathcal{R}$  is this number larger. Here

$$t_c := \min_{\substack{S \subseteq \{1, \dots, n-r\} \\ |S| = \ell}} \sum_{i \in S} h_i, \quad (1)$$

where  $h_i$  is the number of known entries in column  $i$  of  $\mathbf{C}'$ .

*Proof.* Given  $\mathbf{R} \in \mathbb{F}_q^{(r+\ell) \times n}$ , call  $\mathbf{C}'' \subseteq \mathbb{F}_q^{(r+\ell) \times n}$   $\mathbf{R}$ -compatible if  $\mathbf{C}''$  agrees with the known entries of  $\mathbf{C}'$  and  $\text{rowspan}(\mathbf{I} \mid \mathbf{C}'') \subseteq \text{rowspan}(\mathbf{R})$ . We will construct an  $\mathbf{R}$  such that the number of  $\mathbf{R}$ -compatible matrices is

the maximum. By way of illustration, suppose

$$(\mathbf{I} \mid \mathbf{C}') = \left[ \begin{array}{c|ccccc} & 1 & * & 2 & * & * \\ & 0 & * & * & 0 & * \\ & * & 1 & * & * & * \end{array} \right] \in \mathbb{F}_q^{r \times n}$$

where  $*$  are the unknown entries. For any  $\mathbf{R}$ -compatible  $\mathbf{C}''$ ,  $\text{rowspan}(\mathbf{R})$  contains the rows of  $(\mathbf{I} \mid \mathbf{C}'')$  so we may as well assume that these are the first  $r$  rows of  $\mathbf{R}$ . Thus  $\mathbf{R}$  has form

$$\mathbf{R} = \left[ \begin{array}{c|ccccc} & 1 & * & 2 & * & * \\ & 0 & * & * & 0 & * \\ & * & 1 & * & * & * \\ \hline \mathbf{0}_{\ell \times r} & \mathbf{A} & & & & \end{array} \right] \in \mathbb{F}_q^{(r+\ell) \times n}$$

for some  $\mathbf{A} \in \mathbb{F}_q^{\ell \times (n-r)}$  and where we have also assumed the submatrix below  $\mathbf{I}_r$  is  $\mathbf{0}_{\ell \times r}$  since this can always be done with appropriate row operations. Call the top-right submatrix  $\mathbf{B} \in \mathbb{F}_q^{r \times (n-r)}$  so that  $\mathbf{R} = \begin{pmatrix} \mathbf{I} & \mathbf{B} \\ \mathbf{0} & \mathbf{A} \end{pmatrix}$ . To understand how to pick  $\mathbf{A}$  and the  $*$  entries in  $\mathbf{B}$ , let  $\mathbf{b}_i$  be the  $i$ th row of  $\mathbf{B}$  and define  $\mathcal{H}_i$  to be the set of column indices of known entries in row  $i$ . For example, if  $\mathbf{b}'_i = (1 \ * \ 2 \ * \ *)$  then  $\mathcal{H}_i = \{1, 3\}$ . One can observe that the number of  $\mathbf{R}$ -compatible  $\mathbf{C}''$  is equal to the number of ways one can add rows of  $\mathbf{A}$  to  $\mathbf{B}$  without disturbing the known entries. This in turn equals

$$\prod_{i=1}^r |\{\mathbf{x} \in \text{rowspan}(\mathbf{A}) \subseteq \mathbb{F}_q^{n-r} \mid \mathbf{x}_{\mathcal{H}_i} = \mathbf{0}\}| \quad (2)$$

where  $\mathbf{x}_{\mathcal{H}_i}$  means the subvector of  $\mathbf{x}$  restricted to the indices in  $\mathcal{H}_i$ . Note we can assume  $\mathbf{A}$  is full-rank since this can only increase the quantity given in (2). It then follows that the  $i$ th factor equals the cardinality of the left null space of  $\mathbf{A}_{*, \mathcal{H}_i} \in \mathbb{F}_q^{\ell \times |\mathcal{H}_i|}$  and hence equals  $q^{\ell - \text{rank}(\mathbf{A}_{*, \mathcal{H}_i})}$  by rank-nullity. Thus (2) equals  $q^{r\ell - \sum_{i=1}^r \text{rank}(\mathbf{A}_{*, \mathcal{H}_i})}$ .

Hence, the values of  $*$  in  $\mathbf{B}$  do not matter and can be chosen arbitrarily, while  $\mathbf{A} \in \mathbb{F}_q^{\ell \times (n-r)}$  should be chosen full-rank such that the sum  $\sum_{i=1}^r \text{rank}(\mathbf{A}_{*, \mathcal{H}_i})$  is minimized. We can always pick  $\mathbf{A}$  to be all-zero except at some  $\ell$  columns where it equals the identity. This is because, given any (size- $\ell$ ) information set  $\mathcal{I} \subseteq [n-r]$  of  $\mathbf{A}$ , the matrix obtained by zeroing all columns of  $\mathbf{A}$  outside  $\mathcal{I}$  cannot increase this sum. Thus we can pick  $\mathbf{A}_{*, \mathcal{I}} = \mathbf{0}_{\ell \times (n-r-\ell)}$ , and since row operations on  $\mathbf{A}$  do not matter either, also let  $\mathbf{A}_{*, \mathcal{I}} = \mathbf{I}_\ell$ . Thus  $\text{rank}(\mathbf{A}_{*, \mathcal{H}_i}) = |\mathcal{H}_i \cap \mathcal{I}|$  and so  $\sum_{i=1}^r \text{rank}(\mathbf{A}_{*, \mathcal{H}_i}) = \sum_{i=1}^r |\mathcal{H}_i \cap \mathcal{I}|$ . The latter is minimum if  $\mathcal{I}$  indexes the  $\ell$  columns with the fewest known entries. If we denote this number by  $t_c$ , then the number of  $\mathbf{R}$ -compatible matrices is  $q^{r\ell - t_c}$ .  $\square$

The above proposition suggests an optimal strategy to guess an  $(r+\ell)$ -dimensional space such that it contains the row space of a systematic matrix in  $\mathbb{F}_q^{r \times n}$  with partially known entries.

**Theorem 1** (Subspace Guessing Algorithm). *Let  $\mathbf{C} := (\mathbf{I}_r \mid \mathbf{C}') \in \mathbb{F}_q^{r \times n}$  where some entries of the secret matrix  $\mathbf{C}' \in \mathbb{F}_q^{r \times (n-r)}$  are revealed, and let  $\ell > 0$ . Suppose oracle  $\mathcal{O}$  takes as input an  $(r+\ell)$ -dimensional space  $\mathcal{R} \subseteq \mathbb{F}_q^n$  and outputs success if and only if  $\mathcal{R}$  contains the row space of  $(\mathbf{I} \mid \mathbf{C}')$ .*

*Define the following algorithm **GuessSubspace** with oracle access to  $\mathcal{O}$  and which takes the known entries of  $\mathbf{C}'$  as input (i.e., both entry position and entry value).*

1. *Set  $\mathcal{R} := \text{rowspan}(\mathbf{R})$  where  $\mathbf{R} \in \mathbb{F}_q^{(r+\ell) \times n}$  is drawn independently and identically distributed (i.i.d.) across iterations such that*

$$\mathbf{R} = \begin{pmatrix} \mathbf{I}_r & \mathbf{B} \\ \mathbf{0}_{\ell \times r} & \mathbf{A} \end{pmatrix} \in \mathbb{F}_q^{(r+\ell) \times n},$$

where:

- $\mathbf{B} \in \mathbb{F}_q^{r \times (n-r)}$  agrees with  $\mathbf{C}'$  at all known entries and the other entries are sampled uniformly at random, and
- $\mathbf{A} \in \mathbb{F}_q^{\ell \times (n-r)}$  is a fixed matrix such that  $\mathbf{A}_{*,\mathcal{I}} = \mathbf{I}_\ell$  and  $\mathbf{A}_{*,\bar{\mathcal{I}}} = \mathbf{0}_{\ell \times (n-r-\ell)}$  for some (fixed) set  $\mathcal{I} \subseteq [n-r]$  of  $\ell$  columns of  $\mathbf{C}'$  containing the fewest known entries.

2. If  $\mathcal{O}(\mathcal{R})$  returns success, output  $\mathcal{R}$ , else go to Step 1.

If  $\mathbf{C}'$  is drawn uniformly at random, then the above algorithm finds such an  $\mathcal{R}$  in expected number of oracle queries

$$\max\{q^{r(n-r-\ell)-h_c+t_c}, 1\},$$

where  $h_c$  is the total number of known entries in  $\mathbf{C}'$  and  $t_c$  is the number of known entries within the  $\ell$  columns of  $\mathbf{C}'$  that have the fewest known entries (as defined in (1)). Further, this procedure is optimal: no other algorithm with oracle access to  $\mathcal{O}$  can find such an  $\mathbf{R}$  in fewer expected oracle queries (up to a constant factor).

Note that this theorem states that even an adaptive guessing algorithm cannot do better (up to a constant factor).

*Proof.* Call a matrix  $\mathbf{C}'' \in \mathbb{F}_q^{(r+\ell) \times n}$  compatible if it agrees with the known entries of  $\mathbf{C}'$ . There are  $q^{r(n-r)-h}$  compatible matrices, exactly one of which is the correct answer (the matrix  $\mathbf{C}'$ ). Proposition 1 says that any oracle query with an  $(r+\ell)$ -dimensional space  $\mathcal{R} \subseteq \mathbb{F}_q^n$  as input can only check for at most  $q^{r\ell-t_c}$  compatible matrices. Thus, in the worst case, an algorithm has to make at least  $q^{r(n-r)-h}/q^{r\ell-t_c} = q^{r(n-r-\ell)-h+t_c}$  oracle queries, and in the average case, one can improve upon this by only a constant factor of 2 (by checking for each block of  $q^{r\ell-t_c}$  compatible matrices without replacement).

**GuessSubspace** samples  $\mathcal{R}$  as  $\text{rowspan}((\mathbf{I}_r \mid \mathbf{C}'')) + \text{rowspan}((\mathbf{0}_{\ell \times r} \mid \mathbf{A}))$  where  $\mathbf{C}''$  is drawn uniformly from the set of compatible matrices. It is not difficult to show that each distinct  $\mathcal{R}$  is sampled with uniform probability and that each distinct  $\mathcal{R}$  checks for a distinct set of  $q^{r\ell-t_c}$  compatible matrices. In other words, the set of compatible matrices is partitioned by these  $q^{r\ell-t_c}$ -size sets and **GuessSubspace** checks one of them with uniform probability in each iteration. The expected number of oracle queries then follows.  $\square$

Finally, the above results easily extend to *bit-level hints*.

**Corollary 1** (Subspace Guessing with Bit-Level Hints). *Suppose only some bits of some entries of  $\mathbf{C}' \in \mathbb{F}_{2^\nu}^{r \times (n-r)}$  are known. Then Proposition 1 and Theorem 1 remain valid with “known entries” replaced by “known bits”. In other words:*

- There exists an  $(r+\ell)$ -dimensional subspace  $\mathcal{R} \subseteq \mathbb{F}_q^n$  which contains the row space of the largest number of systematic matrices that agree with  $(\mathbf{I}_r \mid \mathbf{C}')$  at the known bits, and this number is  $2^{\nu r \ell - t_c}$ , where  $t_c$  is the number of known bits within any  $\ell$  columns of  $\mathbf{C}'$  with the fewest known bits (so, for example, the maximum value of  $t_c$  is  $\nu r \ell$  instead of  $r \ell$ ).
- Modify **GuessSubspace** in Theorem 1 so that  $\mathbf{A}$  equals the identity on any set of  $\ell$  columns with the fewest known bits. Then the expected number of oracle queries made by this algorithm is  $\max\{2^{\nu r(n-r-\ell)-h_c+t_c}, 1\}$  where  $h_c$  is the total number of known bits. Further, this is optimal.

*Proof.* The proof of Proposition 1 still applies after some modifications. The number of compatible  $\mathbf{C}'$  equals

$$\prod_{i=1}^r |\{\mathbf{x} \in \text{rowspan}(\mathbf{A}) \subseteq \mathbb{F}_{2^\nu}^{n-r} \mid \mathbf{x}_{\mathcal{H}_i} = 0\}|$$

where  $\mathcal{H}_i \subseteq [n-r]$  denotes the known bit positions in row  $i$  and  $\mathbf{x}_{\mathcal{H}_i}$  denotes the restriction of  $\mathbf{x}$  to the bit positions in  $\mathcal{H}_i$ . Since  $\mathbf{x}$  lives in  $\mathbb{F}_{2^\nu}^{n-r}$ , we view  $\mathbf{x}$  as a bitstring in  $\mathbb{F}_2^{\nu(n-r)}$  using a basis of  $\mathbb{F}_{2^\nu}$  over  $\mathbb{F}_2$

in order to interpret  $\mathbf{x}_{\mathcal{H}_i}$ . We can again assume  $\mathbf{A} \in \mathbb{F}_{2^\nu}^{\ell \times (n-r)}$  is zero outside a set  $\mathcal{I} \subseteq [n-r]$  of  $\ell$  column indices and equals  $\mathbf{I}_\ell$  on  $\mathcal{I}$ . Then

$$\begin{aligned} & |\{\mathbf{x} \in \text{rowspan}(\mathbf{A}) \subseteq \mathbb{F}_{2^\nu}^{n-r} \mid \mathbf{x}_{\mathcal{H}_i} = 0\}| \\ &= |\{\mathbf{x} \in \text{rowspan}(\mathbf{A}_{*,\mathcal{I}}) \subseteq \mathbb{F}_{2^\nu}^\ell \mid \mathbf{x}_{\mathcal{H}_i \triangleright \mathcal{I}} = 0\}| \\ &= |\{\mathbf{x} \in \mathbb{F}_{2^\nu}^\ell \mid \mathbf{x}_{\mathcal{H}_i \triangleright \mathcal{I}} = 0\}| \\ &= 2^{\nu\ell - |\mathcal{H}_i \triangleright \mathcal{I}|}, \end{aligned}$$

where we define  $\mathcal{H}_i \triangleright \mathcal{I} \subseteq [\nu(n-r)]$  to be the set of bit positions in  $\mathcal{H}_i$  restricted to the columns in  $\mathcal{I}$ . More formally,  $\mathcal{H}_i \triangleright \mathcal{I} := \{j \in \mathcal{H}_i : \lceil j/\nu \rceil \in \mathcal{I}\}$ . Thus the product equals  $2^{\nu r \ell - \sum_{i=1}^r |\mathcal{H}_i \triangleright \mathcal{I}|}$  and is maximum when  $\mathcal{I}$  consists of the  $\ell$  columns with the fewest known bits. After this, the bit-level version of Theorem 1 follows easily.  $\square$

### 3 Kernel Search with Hints (MinRank)

**Problem 1** (MinRank Problem). *Given  $\mathbf{M}_0, \mathbf{M}_1, \dots, \mathbf{M}_K \in \mathbb{F}_q^{m \times n}$  and a small integer  $r > 0$ , find a nonzero vector  $\mathbf{x} \in \mathbb{F}_q^K$  such that  $\text{rank}(\mathbf{E}(\mathbf{x})) = r$  where  $\mathbf{E}(\mathbf{x}) := \mathbf{M}_0 + \sum_{i=1}^K x_i \mathbf{M}_i$ .*

Let  $\mathbf{E} := \mathbf{E}(\mathbf{x}_0) \in \mathbb{F}_q^{m \times n}$  be the (unknown) matrix of rank  $r$  obtained from a correct solution  $\mathbf{x}_0$  to the problem. Further, assume we can write  $\mathbf{E} = \mathbf{S}(\mathbf{I}_r \mid \mathbf{C}')$  for  $\mathbf{S} \in \mathbb{F}_q^{m \times r}$  and  $\mathbf{C}' \in \mathbb{F}_q^{r \times (n-r)}$ . Suppose we know some hints from both  $\mathbf{S}$  and  $\mathbf{C}'$ . For example, we might know

$$\mathbf{E} = \overbrace{\begin{bmatrix} * & * & * \\ 2 & 1 & * \\ * & 0 & * \\ * & * & * \\ * & * & * \end{bmatrix}}^{\mathbf{S}} \overbrace{\begin{bmatrix} 1 & 0 & 0 & 1 & * & 2 & * & * \\ 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 1 & * & 1 & * & * & * \end{bmatrix}}^{(\mathbf{I} \mid \mathbf{C}')}.$$

We will adapt the kernel search solver [GC00] to exploit these hints. First let us describe the original algorithm.

#### Kernel Search without Hints

Since  $\mathbf{E} \in \mathbb{F}_q^{m \times n}$  has low rank, it has a large kernel. Thus, we can guess vectors  $\mathbf{v}_i \in \mathbb{F}_q^n$  in this kernel and solve the system  $\mathbf{E}(\mathbf{x})\mathbf{v}_i^\top = \mathbf{0} \in \mathbb{F}_q^m$  for  $\mathbf{x} \in \mathbb{F}_q^K$ . The latter yields  $m$  scalar equations, so we need to guess  $b := \lceil \frac{K}{m} \rceil$  many vectors.

1. Guess  $b$  linearly independent vectors  $\mathbf{v}_1, \dots, \mathbf{v}_b \in \mathbb{F}_q^n$  uniformly at random.
2. Solve the system  $\mathbf{E}(\mathbf{x})\mathbf{v}_i^\top = \mathbf{0}^\top \in \mathbb{F}_q^m$  for all  $i = 1, \dots, b$ , for the unknowns  $\mathbf{x} = (x_1, \dots, x_K)$ .
3. If  $\text{rank}(\mathbf{E}(\mathbf{x})) \neq r$ , go to Step 1, else return  $\mathbf{x} \in \mathbb{F}_q^K$ .

#### Kernel Search with Hints

*Hints from  $\mathbf{S}$ .* If we know  $S_{i,j}$  at position  $(i, j)$ , then since  $\mathbf{E}(\mathbf{x}) = (\mathbf{S} \mid \mathbf{S}\mathbf{C}')$ , we get the linear equation  $E(\mathbf{x})_{i,j} = S_{i,j}$  over the unknowns  $x_1, \dots, x_K$ . Thus we can reduce the given problem to a new MinRank instance with  $K-1$  unknowns for each known entry  $S_{i,j}$ , assuming each entry yields linearly independent equations. But the latter is true with high probability for random MinRank instances. Note that this reduction can always be done and is not specific to kernel search.

*Hints from  $\mathbf{C}'$ .* We now adapt the guessing step (Step 1) in kernel search optimally to use the hints in  $\mathbf{C}'$ . Kernel search hopes to draw a  $b$ -dimensional space  $\mathcal{K} \subseteq \mathbb{F}_q^n$  such that it is contained in the  $(n-r)$ -dimensional kernel  $\mathcal{E} \subseteq \mathbb{F}_q^n$  of  $\mathbf{E} \in \mathbb{F}_q^{m \times n}$ . But note  $\mathcal{K} \subseteq \mathcal{E}$  if and only if  $\mathcal{K}^\perp \supseteq \mathcal{E}^\perp$ . Thus we can guess an  $(r+\ell)$ -dimensional space  $K^\perp$  containing the  $r$ -dimensional space  $\mathcal{E}^\perp$  instead, where we set  $\ell = n - r - b$ . Now  $\mathcal{E}^\perp$  is the row space of  $\mathbf{E}$  and, since  $\mathbf{E} = \mathbf{S}(\mathbf{I} \mid \mathbf{C}')$ , also the row space of  $(\mathbf{I} \mid \mathbf{C}')$ . Thus, we can use Theorem 1 to guess  $K^\perp$  optimally, as described next.

Let  $\mathcal{I} \subseteq [n-r]$  be any  $\ell$  columns in  $\mathbf{C}'$  containing the fewest hints. Construct  $\mathbf{A} \in \mathbb{F}_q^{\ell \times (n-r)}$  such that  $\mathbf{A}_{*,\mathcal{I}} = \mathbf{I}_\ell$  and  $\mathbf{A}_{*,\bar{\mathcal{I}}} = \mathbf{0}_{\ell \times (n-r-\ell)}$ .

1. Draw  $\mathbf{B} \in \mathbb{F}_q^{r \times (n-r)}$  uniformly at random such that  $\mathbf{B}$  agrees with  $\mathbf{C}'$  at the hint positions.
2. Let  $\mathbf{v}_1, \dots, \mathbf{v}_b \in \mathbb{F}_q^n$  be a basis of the kernel (i.e., right null space) of  $\mathbf{R} \in \mathbb{F}_q^{(r+\ell) \times n}$  where  $\mathbf{R} := \begin{pmatrix} \mathbf{I}_r & \mathbf{B} \\ \mathbf{0}_{\ell \times r} & \mathbf{A} \end{pmatrix}$ .
3. Solve the system  $\mathbf{E}(\mathbf{x})\mathbf{v}_i^\top = \mathbf{0}^\top \in \mathbb{F}_q^m$  for all  $i = 1, \dots, b$ , for the unknowns  $\mathbf{x} = (x_1, \dots, x_K)$ .
4. If  $\text{rank}(\mathbf{E}(\mathbf{x})) \neq r$ , go to Step 1, else return  $\mathbf{x} \in \mathbb{F}_q^K$ .

### Complexity

The complexity without hints is  $K^\omega q^{r \lceil K/m \rceil}$ . *Derivation:* We succeed exactly when the  $b$ -dimensional space  $\langle \mathbf{v}_1, \dots, \mathbf{v}_b \rangle$  is a subspace of the  $(n-r)$ -dimensional kernel of  $\mathbf{E}$ . This happens with probability  $\binom{n-r}{b}_q / \binom{n}{b}_q \approx q^{-br} = q^{-rK/m}$  where  $\binom{a}{b}_q$  denotes the Gaussian binomial coefficient and where we use  $q^{(a-b)b} \leq \binom{a}{b}_q \leq 4q^{(a-b)b}$  for the approximation. The most expensive steps are solving for  $\mathbf{x}$  in Step 2, costing  $K^\omega$ , and computing the rank of  $\mathbf{E}(\mathbf{x})$ , costing  $mn^{\omega-1}$ . For typical parameters (including Mirath's),  $K > \max\{m, n\}$ , and the first dominates.

The complexity with hints, using Theorem 1, is

$$\tilde{K}^\omega \max\{q^{r \lceil \tilde{K}/m \rceil - h_c + t_c}, 1\}$$

where  $\tilde{K} := \max\{0, K - h_s\}$  and  $h_s$  is the number of hints in  $\mathbf{S}$ ,  $h_c$  is the total number of hints in  $\mathbf{C}'$ , and  $t_c$  is the number of hints within any  $\ell := n - r - \lceil \tilde{K}/m \rceil$  columns of  $\mathbf{C}'$  with the fewest hints. *Derivation:* The hints from  $\mathbf{S}$  allow us to reduce the  $K$  unknowns down to  $K - h_s$ . In the new algorithm, the only additional step in each iteration is Step 2 where we compute a basis of the right null space of  $\mathbf{R}$ . But note that  $\mathbf{R}$  is in row echelon form (since  $\mathbf{A}$  equals the identity at  $\ell$  columns and is zero otherwise) and thus the basis can be directly read off of  $\mathbf{R}$ . Thus, the polynomial cost is the same.

### Bit-level Hints

Suppose  $q = 2^\nu$ . Then, using Corollary 1, one easily extends this algorithm to the case where we know some bits of some entries of  $\mathbf{C}'$  and  $\mathbf{S}$  and not full  $\mathbb{F}_q$ -entries. The complexity becomes  $\tilde{K}^\omega \max\{2^{\nu r \lceil \tilde{K}/m \rceil - h_c + t_c}, 1\}$  where  $\tilde{K} := \max\{0, K - h_s\}$ , and  $h_s$ ,  $h_c$  and  $t_c$  are as before but they count bit-level hints instead.

## 4 GRS with Hints (RSD)

**Problem 2** (Rank Syndrome Decoding (RSD) Problem). *Given  $\mathbf{H} \in \mathbb{F}_{q^m}^{(n-k) \times n}$ ,  $\mathbf{t} \in \mathbb{F}_{q^m}^{n-k}$  and a small integer  $r > 0$ , find a vector  $\mathbf{e} \in \mathbb{F}_{q^m}^n$  such that  $\mathbf{H}\mathbf{e}^\top = \mathbf{t}^\top$  and  $\text{rank}(\mathbf{e}) = r$  (where  $\text{rank}(\mathbf{e})$  is the dimension of the  $\mathbb{F}_q$ -linear space spanned by the entries of  $\mathbf{e}$ ).*



Suppose  $\mathbf{e} \in \mathbb{F}_{q^m}^n$  can be written as  $\mathbf{s}(\mathbf{I}_r|\mathbf{C}')$  where  $\mathbf{s} \in \mathbb{F}_{q^m}^r$  and  $\mathbf{C}' \in \mathbb{F}_q^{r \times (n-r)}$ . Further, let  $\mathbf{E} := \text{ext}(\mathbf{e}) \in \mathbb{F}_q^{m \times n}$  be the matrix obtained by expanding  $\mathbf{e}$  over a fixed basis of  $\mathbb{F}_{q^m}$  over  $\mathbb{F}_q$ . Then we can write  $\mathbf{E} = \mathbf{S}(\mathbf{I}_r|\mathbf{C}')$  where  $\mathbf{S} := \text{ext}(\mathbf{s}) \in \mathbb{F}_q^{m \times r}$ . Our hints will again be entries of  $\mathbf{S}$  and  $\mathbf{C}'$  and we will adapt the GRS solver ([GRS16]) to use them. First let us describe the original algorithm.

### GRS without Hints

Let  $\mathbf{e}$  be decomposed as  $\mathbf{e} = \mathbf{s}\mathbf{C}$ , where  $\mathbf{s} \in \mathbb{F}_{q^m}^r$  and  $\mathbf{C} \in \mathbb{F}_q^{r \times n}$ . Let  $\mathcal{E} \subseteq \mathbb{F}_{q^m}$  be the  $r$ -dimensional space over  $\mathbb{F}_q$  spanned by the entries of  $\mathbf{s}$ . If we guess an  $(r + \ell)$ -dimensional space  $\mathcal{C} \subseteq \mathbb{F}_{q^m}$  such that  $\mathcal{C} \supseteq \mathcal{E}$ , then we can find  $\mathbf{e}$  by solving  $\mathbf{H}\mathbf{e}^\top = \mathbf{t}^\top$ . This holds with high probability provided we have as many equations as unknowns. This approach can be seen as guessing the column space of  $\mathbf{S} := \text{ext}(\mathbf{s}) \in \mathbb{F}_q^{m \times r}$ . Alternatively, we can also guess an  $(r + \ell)$ -dimensional space  $\mathcal{R} \subseteq \mathbb{F}_q^n$  containing the row space of  $\mathbf{C}$ .

We now state the steps for the “row space” version below. The algorithm for the “column space” version is similar.

1. Guess  $r + \ell$  linearly independent vectors  $\mathbf{v}_1, \dots, \mathbf{v}_{r+\ell} \in \mathbb{F}_q^n$  uniformly at random.

2. Write  $\mathbf{e} = \mathbf{a} \begin{pmatrix} \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_{r+\ell} \end{pmatrix}$  for unknown  $\mathbf{a} \in \mathbb{F}_{q^m}^{r+\ell}$  and then substitute it in  $\mathbf{H}\mathbf{e}^\top = \mathbf{t}^\top$  to solve for  $\mathbf{a}$  and hence  $\mathbf{e}$ .

3. If  $\text{rank}(\mathbf{e}) \neq r$ , go to Step 1, else return  $\mathbf{e} \in \mathbb{F}_{q^m}^n$ .

### GRS with Hints

We will describe the modifications for the “row space” version of GRS, assuming  $\mathbf{E} = \mathbf{S}(\mathbf{I}_r|\mathbf{C}') \in \mathbb{F}_q^{m \times n}$ . The “column space” version with hints is analogous, but note it requires the first factor to be systematic instead of the second, i.e.,  $\mathbf{E} = \begin{pmatrix} \mathbf{I}_r \\ \mathbf{S}' \end{pmatrix} \mathbf{C}$  where  $\mathbf{S}' \in \mathbb{F}_q^{(m-r) \times r}$  and  $\mathbf{C} \in \mathbb{F}_q^{r \times n}$  (see Section 6 for details).

*Hints from  $\mathbf{S}$ .* Each known entry  $S_{i,j}$  in  $\mathbf{S}$  gives us an equation of the form  $E_{i,j} = S_{i,j}$  since  $\mathbf{E} = (\mathbf{S}|\mathbf{S}\mathbf{C}')$ . Thus, if we have  $h_s$  hints, then we obtain  $h_s$  additional equations which can lead to a larger value of  $\ell$  in the guessing step.

*Hints from  $\mathbf{C}'$ .* Theorem 1 directly improves the guessing of the  $(r + \ell)$ -dimensional space in Step 1.

### Complexity

The row-space GRS variant with hints yields  $(r + \ell)m$  unknowns and  $(n - k)m + h_s$  equations. Assuming the heuristic they are independent, having at least as many equations as unknowns gives  $\ell \leq n - k - r + \lfloor \frac{h_s}{m} \rfloor$ . Setting  $\ell$  to be its maximum value and using Theorem 1, the complexity is

$$(n - k)^\omega m^\omega \max\{q^{r(k - \lfloor h_s/m \rfloor) - h_c + t_c}, 1\}$$

where  $h_s$  is the number of hints in  $\mathbf{S}$ ,  $t_c$  is the number of hints within any  $\ell = n - k - r + \lfloor h_s/m \rfloor$  columns of  $\mathbf{C}'$  with the fewest hints and  $h_c$  is the total number of hints in  $\mathbf{C}'$ . The polynomial cost is from solving a system with  $(n - k)m$  equations and unknowns. Note since the first  $r$  basis vectors of the  $(r + \ell)$ -dimensional space can be chosen to be systematic (see the matrix  $\mathbf{R}$  in Theorem 1), the additional  $h_s$  equations are of the form  $\text{ext}(\mathbf{a})_{i,j} = E_{i,j} = S_{i,j}$  which can be directly substituted to reduce the unknowns and thus does not increase the polynomial cost.

In the column-space variant of GRS, we guess an  $(r + \ell)$ -dimensional space containing the column span of  $\mathbf{S} \in \mathbb{F}_q^{m \times r}$  (recall here  $\mathbf{S}$  is systematic and not  $\mathbf{C}$ ). We have  $(r + \ell)n$  unknowns and  $(n - k)m + h_c$

equations where  $h_c$  is the number of hints in  $\mathbf{C}$ . We need  $(r + \ell)n \leq (n - k)m + h_c$  so the largest value of  $\ell$  is  $m - r - \lceil (km - h_c)/n \rceil$ . Using Theorem 1, the complexity is

$$(n - k)^\omega m^\omega \max\{q^{r \lceil (km - h_c)/n \rceil - h_s + t_s}, 1\}$$

where  $t_s$  is the number of hints within any  $\ell = m - r - \lceil (km - h_c)/n \rceil$  rows of  $\mathbf{S}'$  with the fewest hints and  $h_s$  is the total number of hints in  $\mathbf{S}'$  where  $\mathbf{S}' \in \mathbb{F}_q^{(m-r) \times r}$  is the non-systematic part of  $\mathbf{S}$ .

### Bit-level Hints

Similar to kernel search, we can also extend these results to bit-level hints using Corollary 1. If  $q = 2^\nu$ , the complexities become

$$(n - k)^\omega m^\omega \max\{2^{\nu r(k - \lfloor h_s/m \rfloor) - h_c + t_c}, 1\}$$

$$\text{and } (n - k)^\omega m^\omega \max\{2^{\nu r \lceil (km - h_c)/n \rceil - h_s + t_s}, 1\}$$

for the row and column space GRS variants respectively.

## 5 Side-Channel Implementation

In this section, we demonstrate how side-channel information can be used to generate hints that assist in solving the MinRank or RSD problems, as discussed in this work. We show how such hints can arise in practice by presenting a side-channel attack that, with a certain success probability, leaks information suitable for constructing these hints. As a target, we focus on the Mirath signature scheme [AAB<sup>+</sup>25], which is based on the MinRank problem.

### A Setup

For our side-channel analysis, we use a ChipWhisperer CW308 setup with a UFO board and an STM32L5HWC as the target device. All measurements are initialized and processed on a host PC, which communicates with the target using the SimpleSerial protocol executed from within a Jupyter Notebook environment.

For the Mirath implementation on the embedded device, we rely on a minimal, functionally relevant subset of the official reference implementation. To the best of our knowledge, no embedded implementation of Mirath currently exists, and a full reference implementation is infeasible on our target device due to runtime memory limitations. This is primarily because of the memory-intensive tree structure used in seed generation.

Our partial implementation focuses on the function `compute_polynomial_proof`, in which the secret key is used.

### B Attack Target

Within the function `emulateMPC_mu`, one of the operations is the computation of  $\mathbf{C}' + \mathbf{C}_{\text{rnd}}$ , where  $\mathbf{C}_{\text{rnd}}$  is a random masking matrix. This computation is performed by calling the function `mirath_matrix_ff_mu_add_mu1ff` (see Listing 1), where the corresponding matrix entries are XORed element-wise. The objective of our attack is to extract the matrix  $\mathbf{C}'$  from this operation. Together with  $\mathbf{S}$ , the matrix  $\mathbf{C}'$  forms the secret key of Mirath.

```

1 static inline void mirath_matrix_ff_mu_add_mu1ff(ff_mu_t *matrix1, const ff_mu_t *matrix2,
2   const ff_t *matrix3, const uint32_t n_rows, const uint32_t n_cols) {
3   for (uint32_t i = 0; i < n_rows; i++) {
4     for (uint32_t j = 0; j < n_cols; j++) {
5       ff_mu_t entry2;
```

```

6         const ff_mu_t entry1 = mirath_matrix_ff_mu_get_entry(matrix2, n_rows, i, j);
7
8         entry2 = mirath_matrix_ff_get_entry(matrix3, n_rows, i, j);
9
10        entry2 = mirath_map_ff_to_ff_mu[entry2];
11        entry2 = entry1 ^ entry2;
12        mirath_matrix_ff_mu_set_entry(matrix1, n_rows, i, j, entry2);
13    }
14 }
15 }

```

Listing 1: Matrix addition in Mirath used in the computation of  $\mathbf{C}' + \mathbf{C}_{\text{rnd}}$ .

The function `mirath_matrix_ff_mu_add_mu1ff` takes as input `matrix2` (corresponding to  $\mathbf{C}_{\text{rnd}}$ ), `matrix3` (corresponding to  $\mathbf{C}'$ ), the number of rows and columns, and `matrix1`, which stores the result  $\mathbf{C}_{\text{aux}}$  of the operation.

For this partial implementation, a random 32-byte seed is generated by a Jupyter Notebook instance and transferred to the target. From this seed, the Mirath public and secret keys are derived as specified by the signature scheme. All remaining values ( $\mathbf{S}_{\text{base}}$ ,  $\mathbf{C}_{\text{base}}$ ,  $\mathbf{v}$ ,  $\mathbf{v}_{\text{base}}$ , and  $\mathbf{\Gamma}$ ) are generated at runtime on the target using the `rand()` function. The random seed for `rand()` is provided by `HAL.GetTick()`, which returns a millisecond-resolution system tick counter.

The program executed on the target performs a full run of `emulateMPC_mu`, and at the end of the execution, the derived matrix  $\mathbf{C}'$  is returned via `SimpleSerial`. The numeric values of the entries of  $\mathbf{C}'$  serve as labels for the side-channel analysis.

Our attacks focus on the parameter set **Mirath-1a-fast** with  $\tau = 17$ ,  $q = 16$ ,  $m = 16$ ,  $n = 16$ ,  $k = 143$ , and  $r = 4$ , where  $\tau$  is the number of times `emulateMPC_mu` is called during signing.

## C Side-Channel Analysis

The attack applied in this work is a profiling side-channel attack. In this setting, we assume that the attacker has access to a device identical to the actual target. On this identical device, a large number of power measurements are recorded while the computation is performed with known secret values. These measurements form the basis for the profiling phase.

Afterwards, a single trace is taken on the target device during a signing operation (or, in our case, during the execution of the partial Mirath implementation). By comparing this trace with the previously recorded measurements from the profiling device, the attacker can get information about which secret key was used during the computation.

### 1) Measurement

The function `emulateMPC_mu` is executed  $\tau$  times during the signing procedure. In this work, we set  $\tau = 17$ , following the **Mirath-1a-fast** parameter set. Since one execution of the target program corresponds to one function call, the target program is called seventeen times using the same seed but with different values for  $\mathbf{C}_{\text{rnd}}$ . During each execution, a power trace is recorded for the function `mirath_matrix_ff_mu_add_mu1ff`, which computes the value of  $\mathbf{C}' + \mathbf{C}_{\text{rnd}}$ . To isolate this section, a trigger is placed immediately before and after the function call.

The seventeen traces obtained from one full execution are combined by computing the mean over all sample points, resulting in a single aggregated trace. Each aggregated trace is associated with a label containing the corresponding entries of the matrix  $\mathbf{C}'$ . In total, 1000 such measurements were recorded to build the dataset used in this work.

## 2) Attack

For the attack phase, a dataset is collected in the same manner as during profiling. The entries of the matrix  $\mathbf{C}'$  are structured as follows: since each element lies in  $\mathbb{F}_{16}$ , four bits are required to represent a single entry, meaning one byte contains two entries. In the **Mirath-1a-fast** parameter set, the matrix  $\mathbf{C}'$  consists of four rows and twelve columns.

In this phase, the measurement traces still contain labels, allowing us to verify whether the classification is correct. The attack proceeds entry by entry, which is feasible because each entry involved in the addition is loaded and processed independently. For the classification of one matrix entry, all profiling traces corresponding to the same value of that entry are grouped together, again by computing the mean of the respective sample points. Thus, for determining the first entry of  $\mathbf{C}'$ , all profiling traces sharing the same value in this position are aggregated, resulting in sixteen different averaged traces—one for each possible field element.

Classification is then performed by computing the correlation between the trace measured on the target device and the grouped profiling traces. Instead of using the full trace, only the relevant portion where the corresponding entry is processed is considered. To achieve this, we define a correlation window consisting of consecutive sample points that contribute most to the classification. The start point of this window corresponds to the sample index with the most significant difference, determined through differences among the profiling groups; for the first entry, this occurs at **start** = 474. From this start point, the window size and shift parameters are computed, with the best results obtained using **window** = 57 and **shift** = 53. Further entries are located using a static column offset **col\_offset** =  $l \cdot 348$ , where  $l$  denotes the column index. If a different row is targeted, the outer loop (see Listing 1) is iterated, and an additional row offset **row\_offset** =  $4208 \cdot k \cdot 2$  is applied, with  $k$  representing the row index.

Moreover, a distinction must be made between the first four bits and the last four bits of each byte, since they correspond to two independent matrix entries. Consequently, two separate classifications are performed, and the start point for the latter case is **start** = 4682. Using these observations, the correlation window boundaries **low** and **high** for each entry can be computed as:

$$\begin{aligned}\text{low} &= \text{start} + \text{row\_offset} + \text{offset} - \text{window} + \text{shift}, \\ \text{high} &= \text{start} + \text{row\_offset} + \text{offset} + \text{window} + \text{shift}.\end{aligned}$$

The correlation itself is calculated using the Pearson correlation coefficient, implemented using the `numpy.corrcoef` function. The class whose correlation window yields the highest coefficient is selected as the predicted value of the corresponding matrix entry.

## D Results of the Attack

We performed a total of 1000 attack measurements and classified them using the method described in Section C. The results show that the individual entries of the secret matrix  $\mathbf{C}'$  can be correctly classified with a success probability of 96.55%. These hints can then be input to the modified kernel search and GRS solvers of Sections 3 and 4 to recover the secret key at a lower cost. Since some hints may be incorrect, the solvers may need to be rerun with the erroneous hint positions being guessed. However, the expected number of retries is small given the high success probability of obtaining the hints.

## 6 Conclusion and Future Work

Our modified kernel search and GRS algorithms (Sections 3 and 4) have the following limitations which are left for future work.

1. We only incorporate hints in kernel search and GRS if one of the matrices in the rank-decomposition of  $\mathbf{E} = \mathbf{S}\mathbf{C}$  is systematic (although we note most cryptographic schemes do this as this reduces key size).

Further, kernel search can be made to guess either the right null space of  $\mathbf{C}$  or the left null space of  $\mathbf{S}$ . Similarly, GRS can guess a space containing the column space of  $\mathbf{S}$  or a space containing the row space of  $\mathbf{C}$ . In both algorithms, both versions have the same complexity when  $m = n$ , the former is better when  $m > n$  and the latter is better when  $n > m$ . However, we can only incorporate hints when the guessing is done for the systematic matrix (although again most cryptographic schemes use  $m = n$  so this would not matter).

Incorporating hints in this case amounts to finding the optimal subspace in Proposition 1 when  $\mathbf{C} \in \mathbb{F}_q^{r \times n}$  is no longer known to be systematic.

2. We adapt only the original GRS algorithm [GRS16], and not the improvement in [AGHT06] which introduces a  $-m$  term in the exponent and thus performs better for certain parameter regimes, including  $m = n$  and  $m < n$ .

Incorporating hints in this case amounts to the following variant of Proposition 1: find an  $(r + \ell)$ -dimensional  $\mathcal{R}$  which maximizes the number of  $\tilde{\mathbf{C}} \in \mathbb{F}_q^{r \times n}$  such that  $\tilde{\mathbf{C}}$  agrees with  $\mathbf{C}$  on the known entries and there exists some polynomial  $p(x) \in \mathbb{F}_q[x]$  such that  $\text{rowspan}(\mathbf{C}p(\mathbf{M})) \subseteq \mathcal{R}$ . Here  $\mathbf{M} \in \mathbb{F}_q^{n \times n}$  is a fixed matrix with an irreducible degree- $n$  minimal polynomial over  $\mathbb{F}_q$ .

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